

Participation Is Not Collaboration: Tracing Public Ownership After Cross-Product Agent Review

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Abstract

Context: A pull request (PR) written by one coding-agent product and reviewed by another looks like a multi-agent workflow. Two products on one PR do not show that one addressed the other. **Objective:** We trace the public path from cross-product review to an addressed edge, a next owner, and a later state. **Method:** We screen 361,296 PRs in a pinned AIDev release and analyse 8,608 trigger PRs, 546 matched pairs, and a 1,067-PR outcome cohort. Exact account, comment, and review-batch identifiers anchor every event to one trigger. We set aside short automation bursts, match cross- to same-product triggers, trace earlier review history, and fix a check point at hour 48. **Results:** Direct dialogue between different products is rare. After a five-minute burst, user accounts own most remaining first actions, and most have reviewed in the repository before. Cross-product triggers get 13 percentage points less visible follow-up than matched same-product triggers. Among PRs still open at that point, an exact reply to the trigger is linked to 17 points more later merge. The link holds under stricter edge rules, a shuffle test inside each repository, and a whole-population model that drops the check point and lets the edge switch on in time. A user account writes that edge four times in five, so it marks human acknowledgement, not agent-to-agent dialogue. **Conclusions:** Public multi-agent impact should be reported as a map of connections, not as a count of products. Addressed edges and visible ownership are useful field signals, but they come from watching, not from an experiment.

Keywords: coding agents, multi-agent software engineering, code review, human-agent collaboration, pull requests, mining software repositories

1 Introduction

On a pull request today you can find two AI coding products at once. One writes the change. A different one reviews it. It looks like two machines working as a team, and it is easy to report it that way.

We followed what happens next, one event at a time. Almost none of it is teamwork. The two products hardly ever answer each other. What we found instead was a person. A maintainer reads the agent’s review point and replies to it. That reply is rare. It is also the one public connection that goes together with the change being merged later. A count of products on a pull request measures none of this.

The gap matters for research and for tool design. A short burst of bot events can look like a conversation. A new review can be part of the same API call. A user account can act next without having been chosen as a go-between. If these cases are mixed together, a product count becomes a weak measure of teamwork.

Field studies already tell us a lot. They count agent-authored PRs, measure how common AI review is, model reviewer order, and study how developers answer agent comments (Li et al. 2026; Selvanayagam and Ghaleb 2026; Zhong et al. 2026a; Cynthia et al. 2026b; Khemissi et al. 2026; Duma et al. 2026). Several report that people still carry the review (Zhong et al. 2026b; Chowdhury et al. 2026). Designed systems show how known agents can pass messages, roles, files, or structured disagreement (Destefanis and Aste 2026; Qiu and Gill 2026; Zhang 2026).

What is still missing is a hard test. These studies count who appears, or they read what was written. Neither tells us whether one actor actually answered another. Our question is narrower and stricter: after one cross-product review event, what connected public path can we point at in the data?

We ask three questions:

RQ1 – Is there a handoff? Once we split one review batch into its parts, and set aside the fast burst of events that follows a trigger, does anything left look like one product handing over to another? And who takes the next visible step? The cost of handing a task between agents has been studied under controlled interruptions (KC and Budathoki 2026). We ask the question that comes before it: can a public handoff be seen at all?

RQ2 – Who bridges the boundary? How much public follow-up happens across the boundary between two products? And when a person steps in, have they worked on this repository before?

RQ3 – Who writes the connection, and does it mark a later state? Two signals say that something connected: a reply pointing at the exact review comment, and automation followed by a person. Does either go together with the change being merged later, and who is it that writes the reply?

We make three contributions.

An evidence ladder with a hard rung. We separate four things that a product count runs together: both products appearing on one PR, a reply that points at the exact review comment, the next visible owner, and the later state of the PR. The second rung is machine-checkable, and it is strict. It is what makes the rest of the paper falsifiable.

A finding that the ladder produces. Each rung deletes most of what the rung above admitted. Almost no cross-product activity becomes one product answering another. The rare exact reply does mark a different later outcome, and a person writes it four times in five. So the measurement choice changes the conclusion, which is the point.

A reusable check for order-based claims. Reviewer order is often read as evidence of coordination. We show that shuffling the order of events inside the same PR reproduces the apparent pattern almost exactly. Prior work on review order compares against an alphabetical baseline (Bouraffa et al. 2025); we argue a shuffled baseline is the one that can falsify a coordination claim.

We do not claim the first observation of AI-to-AI review. Full data rules, all estimates, robustness checks, and rejected experiments are in Online Resource 1 (Supplementary Information).

2 Related Work

2.1 Agent work in public repositories

AIDev maps agent-authored work at GitHub scale (Li et al. 2026). Studies of this and related data examine acceptance, failure, developer roles, merge conflicts, review signals, and cross-PR references (Pinna et al. 2026; Ehsani et al. 2026; Cynthia et al. 2026a; Xu et al. 2026; Nachuma and Zibran 2026; Khemissi et al. 2026). These studies make clear that repository context and human participation matter. They also make merge a poor stand-alone measure of an agent’s quality.

Four lines of work sit closest to ours.

How common it is. The nearest study counts same-product and cross-product AI review, comment output, and speed of reply (Selvanayagam and Ghaleb 2026). It reports that cross-product review is rare and very fast. Our burst windows are a way of taking that speed seriously rather than a new discovery.

Who reviews, and in what order. Zhong et al. model ordered human, LLM, and agent reviewer sequences across review eras (Zhong et al. 2026a). Bouraffa et al. ask whether the order of review activity is meaningful at all, using an alphabetical baseline (Bouraffa et al. 2025). Neither compares the observed order against a shuffled one, which is the check we add.

Who does the work. Several studies find that people still carry the review. Zhong et al. report that developers adopt human suggestions far more often than agent suggestions (Zhong et al. 2026b). Chowdhury et al. report a lower merge rate for PRs reviewed only by an agent than for PRs reviewed only by people (Chowdhury et al. 2026). Duma et al. report that most AI-generated PRs get no review at all (Duma et al. 2026). Our RQ3 result agrees with the direction of this work. What we add is a much stricter definition of what counts as the human contribution, and a design that keeps it before the outcome.

Whether agent feedback helps. Cynthia et al. study how developers respond to individual agent review comments (Cynthia et al. 2026b), and Lin et al. mine developer feedback to a deployed agentic reviewer (Lin et al. 2026). Fatima et al. find that the quality of reviewer-bot feedback shows no clear link to workflow outcomes (Fatima

et al. 2026). That null is worth holding next to our result: we do not measure whether a review comment was good, only whether someone addressed it.

These papers occupy the broad claims that cross-product review exists, that reviewer order matters, and that people answer agent feedback. Our unit is different. It is one cross-product trigger, and the exact edges, batches, owners, and later state that follow it.

2.2 Designed multi-agent review

Role-based systems such as ChatDev and MetaGPT make communication part of the design (Qian et al. 2023; Hong et al. 2023). RepoReviewer divides review into repository, file, priority, and summary roles (Zhang 2026). Adversarial Review adds a critic that checks a review through structured disagreement (Qiu and Gill 2026). SWE-Review tests a known generate-review-revise loop (Wang et al. 2026). Controlled traces can therefore reveal roles and messages directly.

Public GitHub traces are weaker but more natural. Product co-presence does not prove shared memory, a joint plan, or private communication. We use *cross-product feedback* for the trigger and reserve *public coordination* for observable connections and ownership. This measurement boundary is central to our novelty.

3 Data and Method

3.1 Data and event anchor

We pin AIDev-7.6M at revision 37bbe1533e26cc1e1374917dba1186d1c8a4dc81. The release contains 7.69 million agent-authored PRs. We use the richer AIDev-pop layer: 361,296 PRs from repositories with more than 100 stars. Its event tables contain submitted reviews, inline review comments, PR comments, and selected lifecycle events. PRs are created through 31 March 2026; we use 15 April 2026 at 00:00 UTC as the common observation boundary.

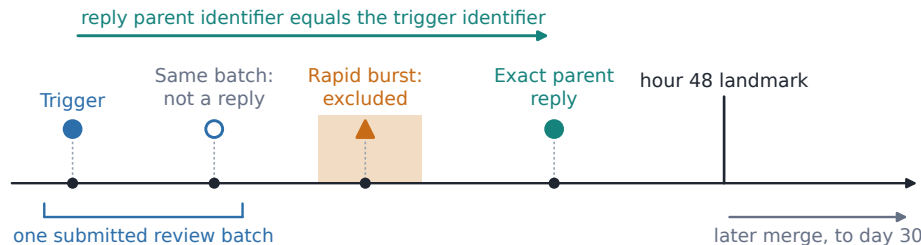
We map only exact official GitHub identities to the six AIDev product labels. Names that merely look similar are not guessed. The PR author carries one product label. A *trigger* is the first review, inline comment, or PR comment written by a different product. We take the earliest such event on the PR, and it must fall after the PR was opened and before it was closed. Everything we measure afterwards is measured from the trigger. Online Resource 1 gives the full identity registry, table coverage, joins, and denominators. Inline comments join to a submitted review by review ID and then to the PR.

An *addressed edge* means one thing only: a later comment carries the trigger comment’s own identifier as its reply target. A reply somewhere else on the same PR does not count. A later review must also carry a different review-batch ID. These rules stop one API call from appearing to answer itself.

We checked what that identifier actually points at, because the paper rests on it. GitHub sets the reply target to the *first* comment of an inline thread, not to whichever comment came just before. The release agrees: across 88,907 inline replies, not one names a comment that is itself a reply, so reply threads are one level deep.

An addressed edge is therefore *a reply in the thread that the trigger opened*. Section 6 reports what that costs us, which turns out to be very little in this cohort. Figure 1 shows the anchor, the three exclusion rules, and the four evidence levels that the rest of the paper keeps separate.

A One trigger, three exclusion rules



B Four evidence levels, read separately

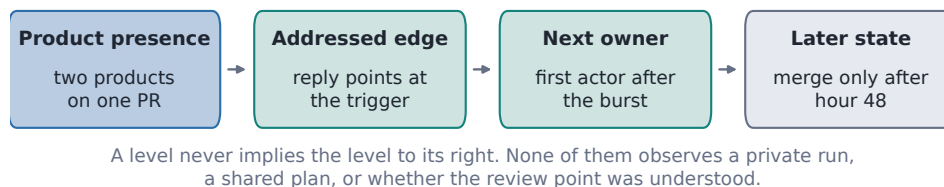


Fig. 1 How we measure. Panel A puts one cross-product trigger on a single PR and marks what each rule removes. A sibling comment in the same submitted review batch is not a reply. An event inside the tested burst window does not define the next owner. Only a comment whose parent identifier is the trigger’s own identifier counts as an addressed edge. The later state is read after a fixed check point at hour 48. Panel B lists the four evidence levels in order. This is a drawing, not data: the positions are for illustration only.

3.2 Three analytic views

A *relay* means the next visible step is taken by a product different from the one that wrote the trigger. It is the thing a multi-agent workflow would produce, and it is what RQ1 looks for.

For RQ1, we follow each first cross-product trigger for seven days, ending at PR closure when earlier. We join exact replies, new review rounds, later PR comments, and visible force-push events. The first later owner is a user account, mapped product, other bot, branch movement or untyped activity, or no action. We repeat this state after excluding the first 0, 1, 5, 10, or 30 minutes. These windows test how much one fast automation burst is driving the answer. They do not reveal a private run boundary. We then apply a second five-minute waiting period, to test whether a mapped-product-first state really becomes a relay to another product.

231 For RQ2 we build matched pairs. Each cross-product trigger is paired with a same-
232 product trigger from the same repository, the same PR-author account, the same
233 author product, the same source channel, and the same month. Pairing uses nearest
234 time without replacement. We also identify the first distinct user account within 48
235 hours. Prior public history requires that account to have submitted a review in the
236 same repository on a different PR strictly before the trigger.

237 For RQ3 we fix a check point at hour 48. We study only PRs that are still open
238 at that point, and we count merges only after it. This keeps cause before effect in
239 time. It is also a limit. Most cross-product inline-trigger PRs finish sooner, so they
240 fall outside the group, and we report how many with the results rather than leaving
241 it unsaid. The group also needs 30 complete days of later observation. The exposure
242 is an exact addressed edge by hour 48; the outcome is merge after hour 48 and by day
243 30. A secondary cohort classifies the first 48 hours as user first, automation then user,
244 automation without a user event, movement, other activity, or no action. This lets us
245 compare an automation-to-user relay with automation alone.

246 We report percentage-point contrasts because they are easy to read. Main intervals
247 cluster or resample by repository. Outcome models adjust for product, channel or
248 month where relevant, trigger age, and public activity known before the trigger. We
249 vary the addressed-edge window, add repository fixed effects, and remove each product
250 pair in turn.

251 The later-state comparison comes from watching, not from an experiment. So we
252 also ask how much hidden structure it would take to remove the result. We run four
253 checks.

254 First, an E-value ([VanderWeele and Ding 2017](#)). It asks a simple question: how
255 strong would a cause we did not measure have to be, before it could wipe out our
256 result? A larger number means a more demanding requirement. Second, a grid that
257 shows how big one missing yes-or-no factor would have to be. Third, three outcomes
258 that had already finished before the trigger. The trigger cannot change the past, so
259 these should show no difference; if one of them does, that is a warning sign and we
260 report it as such. Fourth, a shuffle test that reassigns the exposure label inside each
261 repository.

262 We also run two checks that attack the design rather than the arithmetic. The
263 first drops the hour-48 rule and models the chance of merging over time across the
264 whole trigger population, with the edge switching on when it actually happens. The
265 second splits the edge by whether the person who wrote it had already reviewed in
266 that repository, and then repeats that split inside single repositories.

267 These checks put a limit on fragility. They do not tell us what an intervention
268 would do. Task difficulty and maintainer intent are not measured here. Either one
269 could shape both the public path and the outcome.

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271 3.3 Reproducibility and AI use

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273 The artifact pins the data revision, stores PR-grain cohorts, generates all tables and
274 figures, and runs automated data and leakage checks. We also screened current public
275 review and agent-trace datasets against the same event contract. A released AI-to-AI

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cohort checks product-pair and event-time attribution on shared PRs. A full SWE-
Review-Chat scan applies the exact identity, parent-edge, overlap, and landmark rules.
External rows are not pooled with the main cohort. OpenAI Codex helped inspect
data, write and test analysis code, search literature, make figures, and edit the draft.
The authors checked source records, reran the analysis, verified cited sources, and
reviewed the text and figures. Human authors remain responsible for every claim,
citation, and final decision. No model is an author.

4 Results

4.1 RQ1: product presence rarely becomes a relay

Two products on one PR almost never becomes one product answering the other.
Once the fast automatic burst is set aside, a person usually takes the next visible step.

Figure 2 gives two views of the same narrowing process, over the complete seven-day cohort. Panel A shows how many PRs remain as the connection rule gets stricter. Panel B shows the first visible owner after each tested burst window, among PRs that still have an action. Later public activity is common. An exact reply from a product different from the triggering reviewer is not: it appears on fewer than 1% of PRs. User accounts write four in five of those exact replies.

The burst test changes the owner, not only the time. After five minutes, mapped products lose close to a third of their mapped-first classifications. User accounts then own far more of the remaining actions than mapped products do. This ordering holds in every leave-one-repository-out and leave-one-product-pair-out check.

The next step is even more selective. Among PRs that are still mapped-product first after five minutes, only about one in twenty move next to a different mapped product after a second waiting period.

The same product carrying on is far more common, and it is tempting to read that as an agent staying with the task. It is not. We shuffled the order of the later events inside each PR, keeping that PR's own mix of products, and the shuffled data reproduce the pattern almost exactly. So the trace supports one thing only: a relay to another product is rare. It gives no support for a special product-persistence mechanism.

Answer to RQ1. Two products on one PR greatly overstates how much handing over really happens. Once the fast burst of events is set aside, user accounts are the main visible owner. A relay to another mapped product is rare.

4.2 RQ2: familiar user accounts bridge a quiet boundary

The cross-product boundary is a quieter one. Comparable triggers get less visible follow-up across it, and when someone does step in, it is usually a person who has reviewed in that repository before.

Visible follow-up is less common after a cross-product trigger than after its matched same-product partner. The gap is 13.4 percentage points (Figure 3A). No other outcome supports a claim that this quieter boundary leads to more branch movement or more merges.

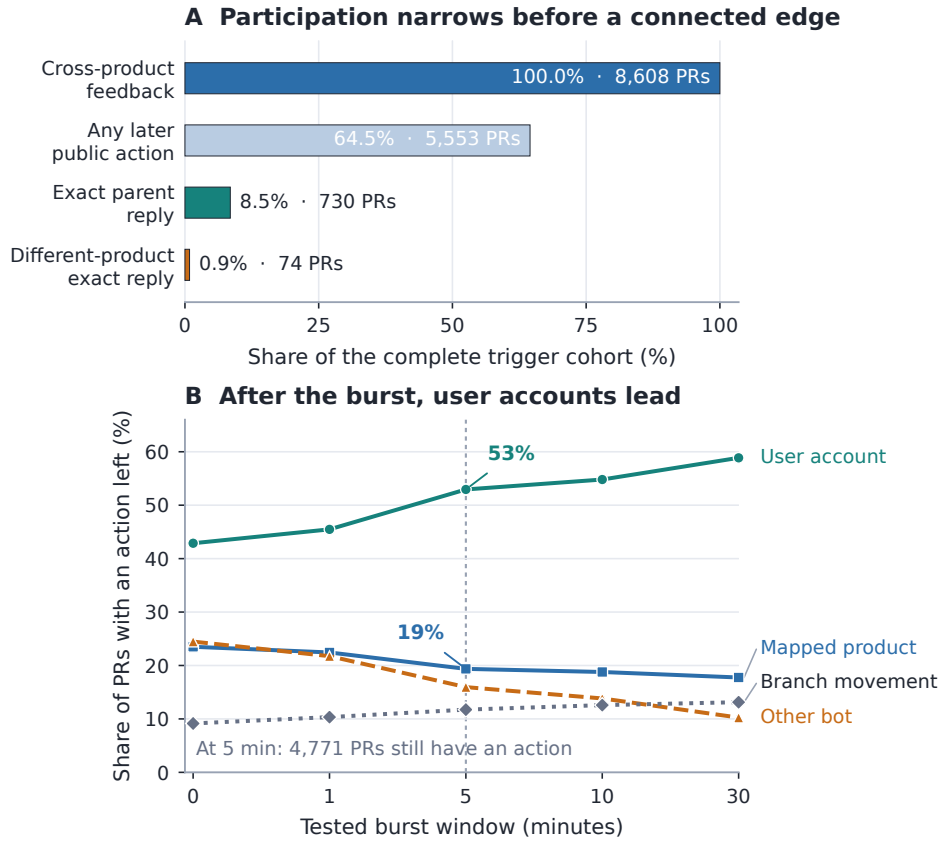


Fig. 2 Public activity narrows before any connected edge appears. Panel A shows the share and count of PRs that meet each stronger evidence rule; the full cohort holds 8,608 PRs. Panel B shows the first visible owner after each of five tested burst windows, among PRs that still have an action. At five minutes, 4,771 PRs still have one, and user accounts own 53% of those actions against 19% for mapped products. The windows are test settings, not known private run boundaries.

Figure 3 joins the matched comparison with a wider history check. Seven in ten of the first user-account bridges had already reviewed another PR in that repository. That rate is close to the rate among all users who answered within 48 hours. So prior history is common right across the response layer. It does not tell us why one account rather than another becomes the bridge.

Answer to RQ2. Comparable cross-product feedback has less public follow-up. When a user account becomes the visible bridge, it is usually not a repository newcomer, but the trace does not show deliberate expert routing.

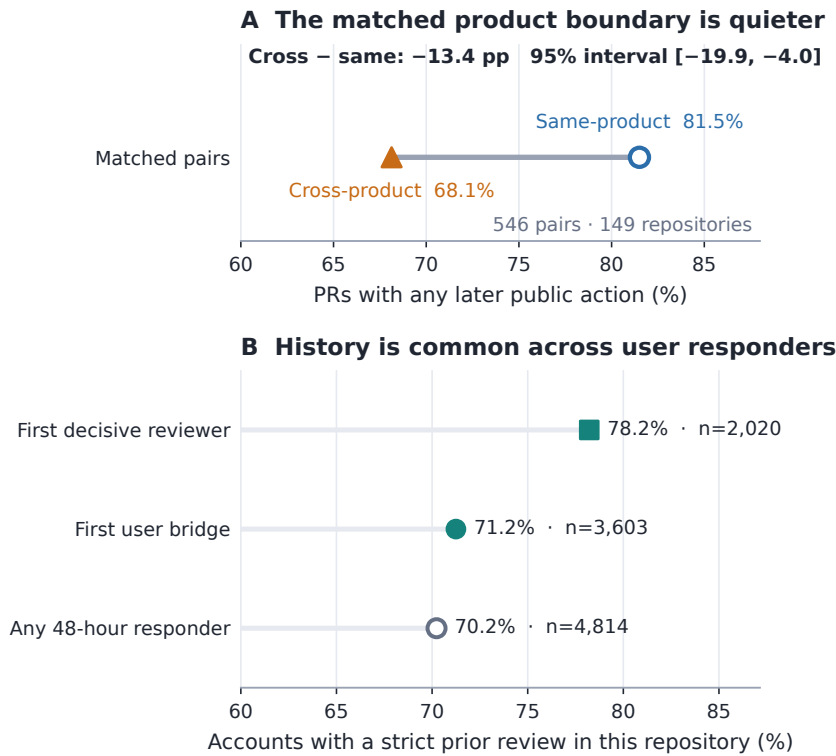


Fig. 3 A quieter boundary, and a history that is common on both sides of it. Panel A compares later public action for cross- and same-product triggers in nearest-time matched pairs. Pairs share repository, author account, author product, source, and month. The sample holds 546 pairs from 149 repositories, with follow-up after 68.1% of cross-product and 81.5% of same-product triggers; the gap is 13.4 points, 95% interval –19.9 to –4.0 from resampling repositories. Panel B compares the first user bridge, the first later decisive reviewer, and all users who answered within 48 hours. A decisive reviewer is one who approves or asks for changes. History means a public review on another PR in the same repository before the trigger. The open circle marks the wider responder baseline. A past review does not tell us whether an account used a tool, remembered the repository, worked by hand, or was picked on purpose.

4.3 RQ3: a person writes the edge that marks a later state

Among the slower PRs that are still open at hour 48, one signal stands out as a marker of later merge: an exact reply to the trigger. Four times in five, a person writes it.

The group holds 1,067 inline-trigger PRs. It is the slower-moving part of the population. Most cross-product inline-trigger PRs close, and most of those merge, at or before hour 48. Our group is the 27% that are still open when the outcome window opens. Everything below describes that slower group, not the most common kind of cross-product PR.

The exposure also needs naming precisely. The rule is a reply whose parent identifier is the trigger’s own identifier. It does not require another product to write that reply. Across the exposed reply events, user accounts write the large majority. The

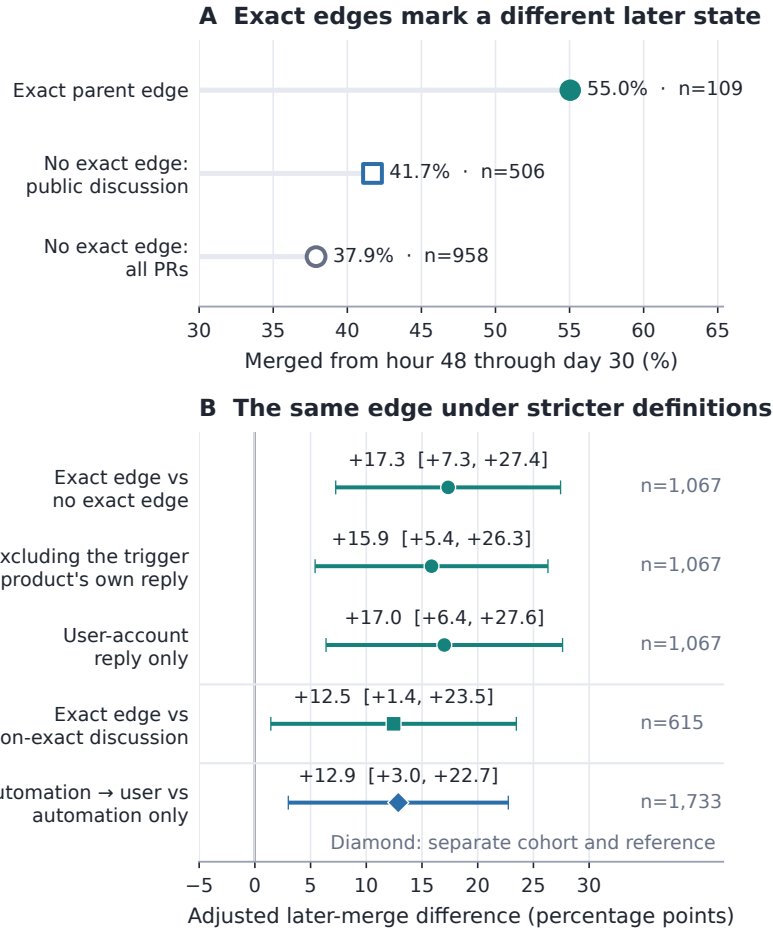


Fig. 4 Connected public signals and later merge. Panel A shows raw later-merge rates: 55% with an exact edge, 41.7% for other public discussion without one, and 37.9% across all PRs without one. The discussion group sits inside the no-edge group. Panel B shows adjusted differences. The top three rows are the same group of PRs under three rules for what counts as an edge: any exact parent reply, the same edge after dropping replies written by the triggering product itself, and edges written by a user account. The fourth row is the discussion-only control. The fifth, marked with a diamond, is the separate automation-to-user route, which has its own group and its own reference. Each row prints its estimate and 95% interval. These are links in observed data, not causes.

triggering product replies to its own comment in a small number of cases, a few more come from bots we do not map to a product, and only three come from a mapped product other than the one that wrote the trigger. So the connected signal here is a person acknowledging an agent's review point. It is not a conversation between agents.

Later merge is more common when there is an exact reply to the trigger than when there is none (Figure 4A). After adjusting for what was already visible before the trigger, the difference is 17.3 percentage points. It stays positive for reply windows

from one to 48 hours, in a model that only compares PRs inside the same repository, and after removing each ordered product pair in turn. We also tested it against a harder comparison. Instead of an edge versus silence, we compared an exact edge with PRs that already carry other public discussion. The result stays positive.

Because most exposed replies are written by people, we refit the model under two stricter rules for what counts as an edge. One drops replies written by the triggering product itself. The other keeps only replies from a user account. Both give the same answer, with intervals that stay clear of zero (Figure 4B, rows two and three). The result is a property of the addressed edge, not of a product answering itself. Only three PRs carry an exact reply from a mapped product other than the trigger's. That is far too few to model. The count is the RQ1 finding showing up again at the outcome stage.

The exact edge is not the only visible thing that goes with a later merge. A wider model that puts all four post-trigger states side by side shows a small group of PRs with branch movement and no discussion sitting even higher, on a very wide interval. Any visible sign that a PR is still being worked on points the same way. We report that model in Online Resource 1 and keep the harder discussion-only comparison as our main specificity check.

The automation-to-user route gives a separate check on a wider group. It is linked to 12.9 points more later merge than automation alone, after the same kind of pre-trigger adjustment. Figure 4 first compares raw later-merge rates, including other public discussion without an exact edge. Panel B shows two exact-edge comparisons and the separate automation-to-user check. The diamond marks the row with a different cohort and reference.

Figure 5 asks how fragile that result is. The E-value here is 2.27. A hidden cause would have to be tied to both the exact edge and to later merge by at least that much, on top of every control we already measure, before it could explain the result away. Panel A shows the same idea as a grid. The shaded corner is where such a factor would have to sit, and it is a demanding corner.

A shuffle test gives a second check, one that rests on the design rather than on a formula. We keep only repositories that hold both kinds of PR. Inside each one we shuffle the exposure label many times and refit. The shuffled results centre on zero, and the real result sits far out in the tail ($p = 0.007$, two-sided). So the finding does not rest on the standard interval formula.

The three before-the-fact outcomes are the weakest of these checks, so we report them as estimates rather than as a verdict. Two of them, an earlier decisive review and earlier branch movement, sit close to zero at every reply window. The third does not. Earlier user activity is +7.0 points at the 48-hour window, with a 95% interval of -1.7 to 15.7. At one hour it is +16.7 points, and that interval clears zero. The one-hour value is a plain failure, and it is why 48 hours is our main window. The 48-hour value is a small leftover, based on too few PRs to be sure. It points in exactly the direction a doubtful reader would expect, so we report it as an estimate and not as a clean zero.

Two further checks matter more than any of the bounds above, because they attack the design rather than the arithmetic.

The first drops the hour-48 rule. Instead of keeping only PRs that are still open, we follow every cross-product inline-trigger PR from its trigger and ask how the chance

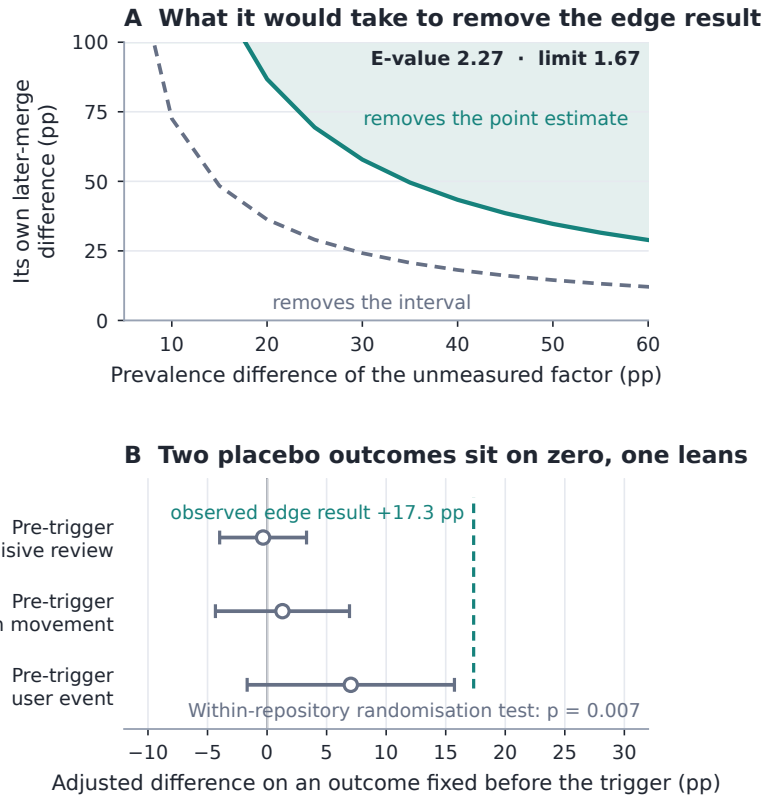


Fig. 5 How much hidden structure it would take to remove the exact-edge result. Panel A shows the arithmetic directly. One missing yes-or-no factor, fixed before the trigger, removes the result only inside the shaded corner. To get there it must be much more common among edge PRs *and* carry a large later-merge difference of its own. The dashed line is the weaker requirement that removes only the interval. As a single summary, the E-value is 2.27, and 1.67 at the interval limit. Panel B applies the same adjusted model to three outcomes that had already finished before the trigger, so the trigger cannot produce them. The teal line marks the observed edge result for scale. Two of the three sit on zero. Earlier user activity leans positive here and clears zero at a one-hour reply window, so the text reports its estimate rather than treating it as a pass. These are limits on fragility, not evidence of cause.

of merging in the next stretch of time changes once an exact reply exists. A PR counts as unexposed until its own reply arrives. The odds of merging are about 1.7 times higher while an edge is active (Figure 6A), and the value barely moves as controls are added. So the result is not an artefact of studying only slower PRs.

The second asks who wrote the edge. We split the exposed PRs by whether the replier had already reviewed in that repository before the trigger. Almost all of the association sits with the familiar repliers. A reply from someone new to the repository shows no positive link at all (Figure 6B). That looks like a mechanism, and we were tempted to report it as one. It does not survive the obvious check. When the comparison is held inside a single repository, the gap between the two kinds of replier

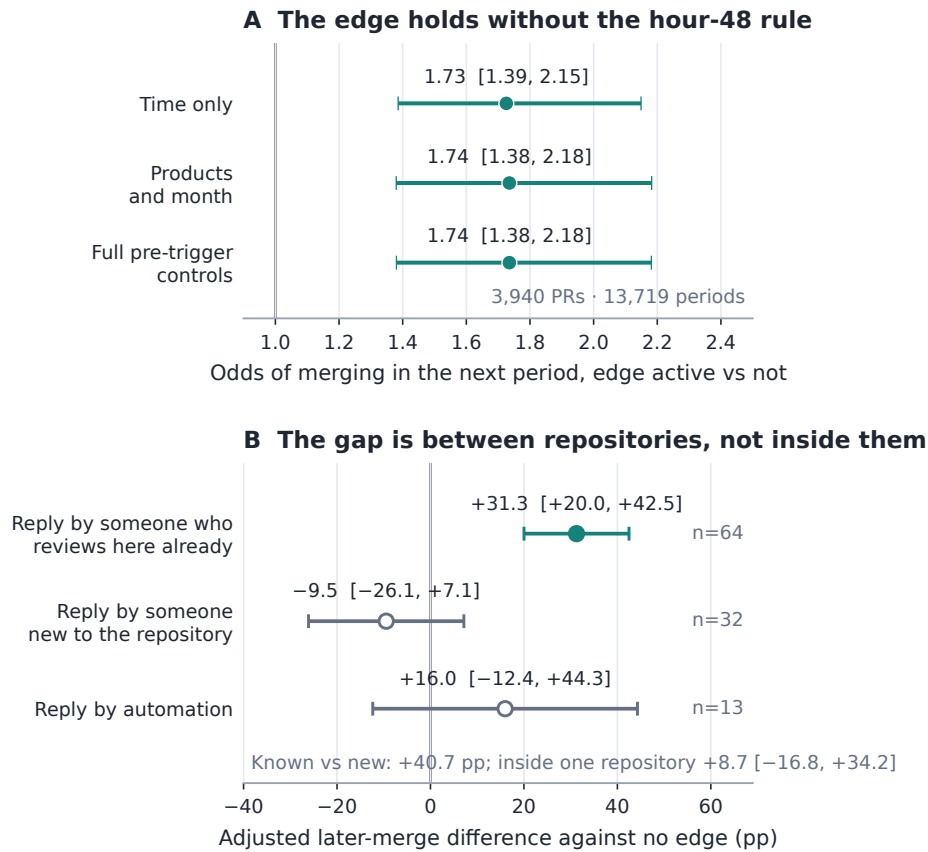


Fig. 6 Two harder tests of the same result. Panel A drops the hour-48 rule entirely. It follows every cross-product inline-trigger PR from its trigger and asks how the chance of merging in the next stretch of time changes once an exact reply exists. A PR counts as unexposed until its own reply arrives, so no time before the reply is credited to it. Panel B splits the edge by who wrote it, against PRs with no edge. The gap between a familiar replier and a newcomer is large, but it nearly disappears when the comparison is held inside one repository, so the trace cannot separate the person from the project.

shrinks to near zero with an interval that spans it. Familiar repliers and newcomers are mostly in *different repositories*, and those repositories differ in how often anything gets merged. We therefore report the split as a description of where the signal sits, not as evidence that knowing the repository is what makes the reply work.

An exact reply may mark a clearer issue, an engaged maintainer, a useful comment, or another unmeasured condition. The data do not tell us which. The result supports the value of a connected trace as a field signal; it does not show that adding a reply would cause a merge.

Answer to RQ3. A person writes the connection, and the connection marks a later state. The exact addressed edge survives every check we ran, including dropping the hour-48 rule and letting the edge switch on in time. Four times in five it is written by a user account rather than by an agent. Nearly all of its association sits with repliers

599 who already review in that repository, but that split is between repositories rather
600 than inside them, so we cannot show that familiarity is the reason. Nothing here is
601 causal.

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603 5 Discussion

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605 5.1 What this changes

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607 The picture a product count paints is wrong in a specific way. It shows two agents in
608 conversation. The trace shows something else: a thin map in which the agents rarely
609 speak to each other, and in which the connection that matters is a person answering
610 an agent. That person is usually not new to the repository. Their reply is rare, it is
611 easy to record, and it is the one public signal in our data that goes together with the
612 change being merged later. This is a link we observed, not a cause we proved. But it
613 points at a different thing to build and a different thing to measure.

614 We also tried, and failed, to name the mechanism. Splitting the edge by whether
615 the replier already reviews in that repository produces a very large gap. Held inside one
616 repository, the gap goes away. The honest reading is that our signal is a composite: it
617 marks a project where people answer review points and PRs get merged, and the trace
618 cannot pull the person and the project apart. That is a limit worth stating plainly,
619 because a reader who saw only the split would draw a stronger conclusion than the
620 data support.

621 Prior work has shown that several reviewer types can appear in one PR, and that
622 their order can relate to review outcomes. Our narrower result explains what that
623 order hides. One review batch can create several events. A short burst can look like
624 a handoff. A product that seems to keep going may just reflect which products are
625 installed. And once a person takes over, the work can bounce back to either product.
626 Exact identifiers let us tell these paths apart.

627 The main field pattern is therefore not a self-running team. It is a thin map of
628 who takes over next. Different-product dialogue is rare, user accounts often bridge the
629 visible boundary, and the most concrete edge is linked to a different later state. This
630 event-level view links natural repository data to the explicit message graphs used in
631 designed multi-agent systems.

632

633 5.2 Implications for agentic software engineering

634

635 **Record the edge, not only the actors.** Review platforms and the platforms that
636 run agents should store four things: what a message replies to, the review-batch ID,
637 the run or session ID, and who is expected to act next. A timestamp on its own cannot
638 separate a burst of parallel events from a real handoff.

639 **Make ownership transfer visible.** A product should be able to mark a review
640 point as received, declined, routed, or awaiting a user. This would turn an inferred
641 event sequence into an auditable coordination contract. The rare different-product
642 relay shows that co-installation alone does not provide this contract.

643 **Use connected signals to guide tests, not as success scores.** The exact
644 edge is a promising marker for checking work, because it is clear and stable. But it is

rare. Only 8.5% of trigger PRs carry one, and about one in ten PRs in the hour-48 group do. It rarely fires by mistake, which makes it useful for auditing. It is not a score, and not yet a quality metric. A prospective study should randomize an explicit acknowledgement or routing feature and then measure resolution, rework, review time, and merge.

Support repository-aware human bridges carefully. Visible user bridges often have prior review history, so systems can offer context from past reviews when asking for help. However, the high baseline among all responders means history should not be sold as proof of expert selection.

5.3 Trying to prove ourselves wrong shaped the story

Several attractive claims did not survive stronger checks. A product that appeared to keep going was matched by a random-order shuffle of the same PR's own events. The newest example is the one above: repository familiarity looked like the reason the edge works, until the comparison was held inside a single repository. Repeated automation did not show a special escalation dose. Post-burst ownership moved in both directions rather than showing one-way user takeover. Prior review history did not support a stable within-repository merge advantage. Request events lacked the target identity needed to prove deliberate routing. Structural comment overlap also remains semantically uncoded. We keep these results and all gates in Online Resource 1 rather than turning them into positive headlines. We also ran the same rules over a large review dataset packaged by other people. After removing the PRs that overlap ours, it left no separate group of exact-edge PRs at the hour-48 check point. That is a failed replication gate, not proof that such edges never occur. It shows why being in the same thread cannot replace a reply that points at the exact comment.

6 Threats to Validity

What we can see. The event tables do not cover the same PRs. A missing row can mean there was no event, or that the event was not captured. Ordinary commit and review timeline rows carry no usable time in this release. So we make claims about timing only for event types that do carry a usable time. We also treat a force-push as branch movement, not as a checked fix.

Identity and meaning. Our list of approved accounts is strict, so it may miss some agent accounts. A GitHub **User** account does not prove that a person worked alone. An exact parent link proves that someone addressed the point. It does not prove understanding, agreement, or repair. We never see private messages, prompts, or shared memory.

Which PRs we kept. The hour-48 check point keeps only PRs that are still open at that moment, and most cross-product inline-trigger PRs close before then. The landmark numbers therefore describe slower PRs. The whole-population model in Figure 6A is our answer to this: it uses every trigger PR, and it still finds the same direction. Shorter reply windows raise a further problem, because a reply inside such a window could itself change whether the PR is still open at hour 48. We treat those windows as extra checks only, and the 48-hour window is the main one.

691 **What the edge means.** An exact parent edge records that someone addressed
692 the trigger. It does not record who. In this group that someone is almost always a user
693 account, so RQ3 measures a person acknowledging an agent, not one product handing
694 over to another.

695 The reply identifier is looser than the name suggests. It points at the first comment
696 of a thread, so our edge means “a reply in the thread the trigger opened”. Two things
697 follow, and we measured both. First, the exposure can only fire when the trigger
698 is itself the first comment of its thread. Among cross-product inline triggers that is
699 almost always true: 13 of 3,526 sit mid-thread, and in the hour-48 group only 3 of
700 1,067 do, none of them exposed. The restriction bites hard for same-product triggers,
701 where 46% sit mid-thread, which is one more reason we do not read the same-product
702 edge rate as comparable. Second, inside a thread with several replies, a later reply may
703 be answering an earlier reply rather than the trigger. Among the 109 exposed PRs,
704 92 have exactly one reply in the thread, so for those the reply can only be answering
705 the trigger. For the remaining 17 we cannot tell.

706 **Selection and cause.** Matching holds several visible factors fixed, and the out-
707 come models use pre-trigger controls and repository-aware intervals. They cannot
708 balance what a review meant, how hard the task was, what the maintainer intended,
709 or what happened in private.

710 The stress test puts a limit on this. A hidden cause weaker than an E-value of 2.27
711 cannot account for the edge result. A stronger one still could. And the before-the-
712 fact user-activity check shows that selection of this kind is present in our data, and
713 only partly measured. Our controls also include no stand-in for task difficulty. The
714 later-state findings stay associations.

715 **How far the results reach.** AIDev-pop covers public repositories with more
716 than 100 stars, and the product pairs are uneven. Leave-one-out checks ease the worry
717 that a few repositories or a few pairs drive the result. They do not extend it to private
718 projects or to future versions of these products. The outside dataset we screened checks
719 our schema and identities, but after removing overlap it gives no independent group
720 for RQ3.

721

722 7 Conclusion

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724 Two AI products on one pull request is not proof of teamwork. Once the fast burst of
725 events is set aside, product co-presence almost never becomes one product answering
726 another. User accounts form the main visible bridge, and most of them have reviewed
727 in that repository before.

728 The clearest connected signal is an exact reply to the trigger. It marks more later
729 merges, and it survives stricter edge rules, a shuffle test, and a model that drops the
730 hour-48 rule and uses the whole population. A person writes that signal four times in
731 five. So the edge that carries the weight is a human answering an agent, and that is
732 exactly what a count of products does not measure. Field evaluation of multi-agent
733 work should ask for three things: an addressed edge, a visible owner, and a later state.

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Statements and Declarations	737
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